

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

If n is the number of irrational terms in the expansion of $\left(3^{1/4} + 5^{1/8}\right)^{60}$, then $(n - 1)$ is divisible by:

- (1) 26
- (2) 30
- (3) 8
- (4) 7

Q2: 16 March (Shift 1) - Single Correct

Let $[x]$ denote greatest integer less than or equal to x . If for $n \in \mathbb{N}$, $(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$, then $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j} + 1$ is equal to :

- (1) 2
- (2) 2^{n-1}
- (3) 1
- (4) n

Q3: 16 March (Shift 2) - Numerical

Let n be a positive integer. Let

$$A = \sum_{k=0}^n (-1)^k n_{C_k} \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k \right]$$

If $63A = 1 - \frac{1}{2^{30}}$, then n is equal to _____

Q4: 17 March (Shift 1) - Single Correct

If the fourth term in the expansion of $(x + x^{\log_2 x})^7$ is 4480, then the value of x where $x \in \mathbb{N}$ is equal to :

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- (1) 2
- (2) 4
- (3) 3
- (4) 1

Q5: 17 March (Shift 1) - Numerical

If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

Q6: 17 March (Shift 2) - Single Correct

The value of $\sum_{r=0}^6 ({}^6C_r - {}^6C_{6-r})$ is equal to :

- (1) 1124
- (2) 1324
- (3) 1024
- (4) 924

Q7: 17 March (Shift 2) - Numerical

Let the coefficients of third, fourth and fifth terms in the expansion of $\left(x + \frac{a}{x^2}\right)^n$, $x \neq 0$, be in the ratio 12: 8: 3. Then the term independent of x in the expansion, is equal to _____.

Q8: 18 March (Shift 1) - Single Correct

Let $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + \dots + a_{40}x^{40}$

then $a_1 + a_3 + a_5 + \dots + a_{37}$ is equal to

- (1) $2^{20} (2^{20} - 21)$
- (2) $2^{19} (2^{20} - 21)$
- (3) $2^{19} (2^{20} + 21)$
- (4) $2^{20} (2^{20} + 21)$

Q9: 18 March (Shift 2) - Numerical

The term independent of x in the expansion of

$$\left[\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}} \right]^{10}, x \neq 1, \text{ is equal to } \underline{\hspace{2cm}}$$

Q10: 18 March (Shift 2) - Numerical

Let nC_r denote the binomial coefficient of x^r in the expansion of $(1+x)^n$.

$$\text{If } \sum_{k=0}^{10} (2^2 + 3k) {}^nC_k = \alpha \cdot 3^{10} + \beta \cdot 2^{10}, \alpha, \beta \in R,$$

then $\alpha + \beta$ is equal to _____

Q11: 18 March (Shift 2) - Numerical

Let P be a plane containing the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+5}{2}$ and parallel to the line $\frac{x-3}{4} = \frac{y-2}{-3} = \frac{z+5}{7}$. If the point $(1, -1, \alpha)$ lies on the plane P , then the value of $|5\alpha|$ is equal to _____

Answer Key**Q1** (1)**Q2** (3)**Q3** (6)**Q4** (1)**Q5** (4)**Q6** (4)**Q7** (4)**Q8** (2)**Q9** (210)**Q10** (19)**Q11** (38)